Aminu Daniel Ananyan

Software Engineering Student | Blockchain & Full-Stack Developer

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SUMMARY

Software Engineering undergraduate with hands-on experience in applied cryptography, blockchain application development, and full-stack web and mobile engineering. Built and defended six technical projects during a Bitnob (fintech) internship, including a deployed Ethereum dashboard and a from-scratch PKCS#7/AES-CBC implementation. Comfortable working from primary sources (specifications, standards, and papers) rather than second-hand summaries.

EDUCATION

B.Sc., Software Engineering, Nile University of Nigeria, Abuja. Expected 2028 · Faculty of Computing

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Dart

Frontend: React, Next.js, Vite, HTML, CSS

Mobile & Desktop: Flutter

Blockchain: Ethereum (Sepolia), ethers.js v6, Alchemy JSON-RPC, event log decoding, ABI encoding

Cryptography: AES-CBC, PKCS#7 padding, DES/3DES, classical cipher cryptanalysis (frequency analysis, hill-climbing), PyCryptodome

Testing: pytest

Tools & Platforms: Git, GitHub, Cursor IDE, Claude Code, Figma, Vercel

EXPERIENCE

Software Engineering Intern (SIWES), Bitnob (subsidiary of Mainhedge Inc.), Abuja *July 2026 – September 2026*

Reported to Olayinka Omoniyi (Software Engineer). Bitnob is a cross-border payments company operating on Bitcoin and stablecoin rails, with >$4.6B in processed volume and ISO 9001, ISO 27001, and SOC 2 Type II certification.

- Completed and defended six technical projects in applied cryptography and blockchain application development, presented weekly to an engineering supervisor.
- Built **ContractLens**, a React/TypeScript dashboard that decodes raw Ethereum event logs into readable smart contract activity, deployed to production.
- Implemented **PKCS#7 padding and AES-CBC encryption** from first principles in Python, with a pytest suite proving both correct behaviour and the presence of two deliberately introduced bugs (off-by-one unpad, silent corruption).
- Authored technical case studies on Kerckhoffs's principle, modern stream ciphers (RC4, Salsa20, ChaCha20), and a comparative performance benchmark of 3DES vs. AES demonstrating the structural reasons for AES's ~10× throughput advantage.
- Participated in code reviews on the Bitnob application, identifying optimisation and security improvements.

SELECTED PROJECTS

ContractLens: Ethereum Smart Contract Activity Dashboard *github.com/DanDeeLegend/ContractLens*

React, TypeScript, Vite, ethers.js v6, Alchemy, React Router

- Solves the "public but not accessible" problem of on-chain data: given a contract address, retrieves event logs from Ethereum Sepolia and decodes them into a readable activity feed (event type, block, transaction hash, participating addresses).
- Matches Keccak-256 event-signature hashes against known signatures; decodes indexed parameters from topics and non-indexed parameters from ABI-encoded data.
- Positioned as a lightweight Etherscan alternative for single-contract monitoring.
- Deliberate architectural choice to depend on Alchemy JSON-RPC rather than run a node, documented in the project write-up.

Padding Scheme Explorer: Cryptographic Correctness Demonstrator *github.com/DanDeeLegend/padding-scheme-explorer*

Python, PyCryptodome, pytest

- From-scratch PKCS#7 padding implementation integrated with real AES-CBC encryption.
- Ships a correct implementation alongside two deliberately buggy variants, with tests proving the behaviour of each.
- Demonstrates the padding-oracle attack class by making padding-validity distinguishable, the exact signal an attacker exploits.

SignalMap: Campus Network Signal Quality Mapping

Flutter, Figma, Google Forms

- Application for mapping WiFi and cellular signal quality across a university campus.
- Analysed 60+ Google Form survey responses on campus network disruptions using pivot-table analysis.
- Implemented a pixel-accurate Flutter UI from a Figma export, extracting design tokens and building the widget tree end-to-end.
- Set up a Docs-as-Code project structure (PRD, SRS, architecture docs) with Git hygiene around raw survey data.

The Board: Personal Life-Management App *in development*

Flutter, Dart

- Original product concept: business-management metaphor applied to individual life management. Measures drift from the user's own baseline rather than fixed goals; hybrid passive + prompted data capture.
- Phase 1 complete: pure-Dart domain model and drift-detection engine with test coverage.
- Target first users: students on my university campus.

Bright Spot: Local-First Password Manager *in planning*

Android, Flutter (planned)

- Designing an unconventional, local-first password manager for an individual end-user, distributed as a sideloaded Android APK.
- Core UX decision: device-to-device migration via QR code scan rather than cloud sync.
- Architecture and product spec complete; implementation not yet started.

WRITING

- **Medium** (medium.com/@danielaminu2): technical articles on engineering topics.
- **Substack** (substack.com/@dannii0): essays on personal growth, faith, and reflection.

INTERESTS

Frontend design aesthetics · AI engineering · Blockchain and cryptography · Technical writing · Building products for real users